

Conglomerate Mergers: Effects on Growth and Competition

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Abstract

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1 Introduction

Conglomerative mergers are important to many fields of economic inquiry, ranging from industrial organization [3, 11, 12], international trade [4, 22, 15], to finance [14, 5] and even management studies [2, 1]. However, unlike horizontal or vertical mergers, whose drivers have been studied thoroughly, the determinants of conglomerative mergers have not received nearly as much attention. Formally, it defines the merging of two businesses that operate in different markets and are not part of the same value chain. Many of these mergers take place across borders, which is why they are also known as complex foreign direct investment [4]. For the purpose of this article, we will look at *pure* conglomeration, omitting motivations of market or product extension [10]. The determinants of such processes remain at once manifold and elusive. Among those brought forward are economies of scale, and increasing market power [17], taxation [8] or alignment with management salary risk preferences [1]. One overarching theme, however, is the reduction in risk. For example, [25] points to a reduction in the standard deviation of profits by merging two firms operating in independent markets. Despite the fact that the standard deviation is neither technically, nor methodologically a good measure of risk [29], it provides valuable insights into the determinants of conglomerative mergers. In this context [17] introduces the concept of growth maximization as a reason for conglomeration, where diversifying across firms yields an increased rate of growth for the owning entity. Considering this in light of recent ideas by [19] offers a new path forward. In particular, the notion of firm growth rate optimization dynamically through time through the reduction of losses seems both novel and established. Under this framework an agent (here a firm) searches for the most cost-effective means to reduce negative impacts on their growth rate. [1] propose that mergers act to avoid disastrous outcomes, almost as some form of insurance, provided that the premium paid in terms of transaction costs is low enough. Adding more companies to a portfolio to reduce losses works best, of course, when these companies' outcomes are maximally independent. [3] presents this thought by proposing that mergers happen at different stages of the

firms' lifecycle, thus increasing the diversification potential.

Just as unsolved remains question of the impact that such mergers have on the competitive landscape. While [32] finds that post acquisition companies grow at an above industry average rate, [12] and [11] do not detect any consequences that would increase market concentration. The results in [32] seem to depend on the acquiring company to financially support the acquired company as a necessary condition of out performance. Naturally, there is a limit to the size of a company as transaction cost, such as the coordination of all constituent businesses, increase [7]. This provides a natural drag on the effectiveness of conglomeration as a means to increase the growth rate. The direction of the effect on competitiveness is of course relevant to policy-makers, wishing to optimize consumer welfare.

We will offer evidence to help propel the answer to both of these questions. Section 2 provides an introduction into growth rate optimization under multiplicative dynamics and their relevance in trying to understand competitive processes. Section 3 presents the formal model, as well as the assumptions implied in it. Continuing in Section 4, we show the results of the simulation, and introduce and discuss the metrics used in our arguments. We discuss these in Section 5 before concluding in Section 6.

2 Background

Conventional economic models had great success explaining the formation of vertical and horizontal mergers. Despite this, conglomerative mergers have escaped such reasoning so far. Many explanations are based on cost sharing arguments [17], whereby firms split up expenses among themselves. One point often made against such an explanation is that this is in effect zero-sum, given that one firm's gain is another's loss. However, this critique only holds under the assumption of additive growth. In a setting of multiplicative dynamics, where the next state s_{t+1} of some variable (e.g. market share, reputation, size) is dependent on the previous one s_t , the objective is to maximize the growth rate g , the geometric average of periodic increments $\delta_t \in (0, \infty)$, sometimes also referred to as the Compound Annual Growth

Rate in investment circles

$$g = \left(\prod_{t=1}^T \delta_t \right)^{\frac{1}{T}} \quad (1)$$

These increments are the changes to the variables in question, such as the change in sales, the effect of marketing campaigns on the way consumers perceive as firm, etc. Crucially, the lower increments are, the greater their impact on the growth rate. While an increment of $\delta_t = 0.2$ already requires the next one to be $\delta_{t+1} = 1.25$ to compensate it, one of $\delta_t = 0.5$ requires δ_{t+1} to be 2. This asymmetry in influence of losses over gains grows exponentially and thus follows a logarithmic function of the increments on the growth rate as seen in Figure 1. From the point of view of avoiding the smaller levels of δ_t or increasing them across the board, sharing cost and even revenues in a conglomerate becomes a positive-sum game, as the reduction in the gain of the more successful firms is overcompensated by the increase in gain of the others. This holds true regardless of whether all firms make profits, only some do, or even none do. Suppose there are two firms in a conglomerate and absolute increments defined as $\Delta_{it} = \delta_{it}s_{it} - s_{it}$. Furthermore, without loss of generality suppose $\Delta_{it} > \Delta_{jt}$ and a sharing level of x . Given that $\frac{\partial^2 \log(x)}{\partial^2 x} < 0$, it is clear that $\log(\Delta_{it}) - \log(\Delta_{it} - x) < \log(\Delta_{jt} + x) - \log(\Delta_{jt})$.

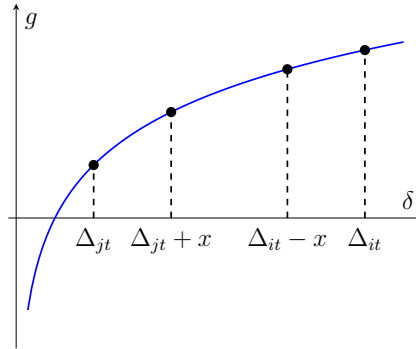


Figure 1: Impact of growth increments on the compound growth rate.

One can think of this as raising the base of the state s_{t+1} such that the

next growth increment δ_{it+1} impacts a greater absolute value. For example, assuming a multiplicative process consisting of normally distributed increments with mean μ and standard deviation σ , the time average growth rate, the average increment, converges to

$$g = \mu - \frac{\sigma^2}{2} \quad (2)$$

In addition to the mean μ , the fluctuation of the increments $\frac{\sigma^2}{2}$ has an outsized impact on the growth rate, as the negative fluctuations outweigh the positive ones even though they are symmetric.¹ This dynamic is sometimes referred to as *variance drain* [16] or *volatility drag* [28]. Thus, as outlined, reducing the magnitude of fluctuations is the largest lever to increase the growth rate.

Again, we can link this to sharing, where conglomerate formation increases g for all firms within the entity. Under multiplicative dynamics, sharing can be shown to be growth optimal in the following way [20]. With N firms, whose growth increments δ_i are independently drawn from the same distribution $\mathcal{N} \sim (\mu, \sigma)$, the *sharing* or *pooling* of their increments increases their individual growth rates such that

$$g = \mu - \frac{\sigma^2}{2N} \quad (3)$$

As the number of firms N in a conglomerate increases, the drag imposed by the fluctuations decreases. In the extreme case of $N \rightarrow \infty$ each firm is able to realize the ensemble average μ with every increment. Of course there are limits to how big a firm and thus a conglomerate can grow, given that transaction cost, especially for coordination increase [7]. Any increase in conglomerate size will increase transaction cost, while at the same time increasing the growth rate for all firms included. Thus, one can see this kind of cooperation as a hedge, or an insurance against unfavorable growth incre-

¹ $(1 - \delta_t)(1 + \delta_t) = 1 - \delta_t^2 < 1.$

ments, the premium for which is the cost of coordination. Clearly, there is a trade-off involved, and a conglomerate size, though certainly varying, which optimizes it. This concept is somewhat related to that of diversification, as applied to financial portfolios. However, in contrast to the latter, pooling also improves the growth rates of the entities that make up the portfolio, while offering even greater growth prospects than diversification to the portfolio itself. It is in this sense, *true diversification*.

3 Model

3.1 Random Multiplicative Growth

3.1.1 Growth Process

To investigate how this influences the incentives in conglomerate formation, we set up an Agent Based Model with M markets that are uncorrelated between each other, each containing an amount of N firms. Firms face random multiplicative growth that is unique to their market. Each increment δ_{it} is drawn from a normal distribution $\mathcal{N}(\mu_m, \sigma_m)$ with positive ensemble average μ_m . In this framework each firm's periodic state, which can be seen as its size, is multiplicatively linked to its previous one.

$$s_{it+1} = s_{it}\delta_{it} \quad \text{with} \quad \delta_{it} \sim \mathcal{N}(\mu_m, \sigma_m) \quad (4)$$

We model markets such that the mean and the standard deviation of the distribution over their growth increments are positively correlated, meaning that a higher mean μ_m comes with a higher standard deviation σ_m . As explained in section 2, in multiplicative processes one is not just maximizing for the ensemble average μ_m , but for the growth rate g . Hence, a higher scale σ_m decreases the growth rate, through the volatility tax.

Each firm thus realizes an absolute period gain of $\Delta_{it} = s_{it-1}\delta_{it} - s_{it-1}$. For any firm not part of a conglomerate this will also be their period profit $\Pi_{it} = \Delta_{it}$. Firms in conglomerates C pool an amount $\alpha\Delta_{it}$, which flows into

the conglomerate's period pool $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it}$. However, as conglomerates grow in size, maintaining them becomes ever more difficult as transaction costs increase. We model these as a logistic function of the form

$$\Phi(O_c, M) = \frac{1}{1 + \exp^{-|O_c| + 0.4M}} \quad (5)$$

This factor scales the cost according to conglomerate size $\sum_{i \in c} s_{it}$ such that the pool size is given by

$$\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it} \quad (6)$$

where the set of markets a conglomerate operates in is defined as $O_c \subseteq M$. Notice that the factor 0.4 in the logistic function scales the costs such that a conglomerate operating in 40 % of all markets would lose 50 % of their market value in transaction cost every round. Lastly, this pool will be shared equally, determining a member's profit by $\Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_{c_i}|}$. Consequently, any firm's next state is given by $s_{it+1} = s_{it} + \Pi_{it}$.

3.1.2 Merging and Splitting

At the beginning of every time step t each firm $i \in M \times N$ gets a chance to choose another with probability p . If successful, it will select a random other firm j , among the remaining $(M \times N) - 1$ firms. We allow conglomerates to only operate one firm per market, meaning it can have a maximum of M members. Hence, conglomeration is only *feasible* if firm j and i :

1. don't operate in the same market ($m_i \neq m_j$)
2. are not part of a conglomerate that operates in a market in which the other firm, or the conglomerate which it is part of operates in ($O_{c_i} \cap O_{c_j} = \emptyset$)

We assume that if both firms are in conglomerates that these will merge as

well. After confirming *feasibility*, merging firms determine *desirability*, meaning that they would both benefit from joining forces. They do so by comparing the realized pool Ω size over the last l periods, against a “synthetic” pool $\hat{\Omega}$ size if they had merged. If $\hat{\Omega} > \Omega$ the merger will be implemented. One could think of this as some form of due diligence.

After an initial l periods each firm that is part of a conglomerate evaluates its performance within it by comparing its growth rate $g_{i(t-l):t}$ being part of the conglomerate against the potential rate $\hat{g}_{i(t-l):t}$ it had realized, had it remained outside. Upon being worse off, a firm exits the conglomerate and continues to operate alone. The full model can be seen in algorithm 1.

Algorithm 1: Model flow with random growth.

```

for  $t$  in  $T$  do
  foreach  $i \in N \times M$  do
     $\lfloor$  Merge with  $j$  if  $(m_i \neq m_j)$  and  $(O_{c_i} \cap O_{c_j} = \emptyset)$  and  $\hat{\Omega} > \Omega$ 
  foreach  $c \in C$  do
     $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it}$ 
    foreach  $i \in c$  do
       $\lfloor$   $\Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_c|}$ 
    foreach  $i \notin c$  for all  $c \in C$  do
       $\lfloor$   $\Pi_{it} = \Delta_{it}$ 
    foreach  $i \in c$  for all  $c \in C$  do
       $\lfloor$  Exit  $c$  if  $\hat{g}_{i(t-l):t} > g_{i(t-l):t}$ 

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4 Results

We implemented the models outlined in the previous section in python with a parameter specification that can be found in Table 1. Note that we conducted extensive robustness testing, none of which altered the central findings of this section.

To answer the questions outlined in Section 2, we analyze certain outputs of our model. We gradually increase the parameter of interest, the sharing

Parameter	Value
M	100
N	100
T	1000
l	50
p	0.05
α	$[0, 0.5]$

Table 1: Model Parameters and their values

rate α , by 0.02 to investigate the behavior of our model. To gain further confidence in our result, we ran each experimental setup (each level of α) 10 times in a Monte Carlo type setting and averaged them.

Firstly, Figure 2 indicates a stable tendency to form conglomerates, which after initial fluctuations, partially due the review period l , settles to a stable level up until the final number of iterations T is reached. Note that we omit the no-sharing case in this figure. Both the number of conglomerates, and their size converge to a regular pattern. Interestingly, while initially increasing the sharing rate leads to larger conglomerates, there seems to be an optimum sharing level between 0.2 and 0.3 that maximizes. This is also reflected, though not as strongly, in the number of conglomerates, which bottoms out at the same level. This is not a surprising result, given that one would expect an inverse relationship between the size and the number of conglomerates. Both of these findings, lend support to the idea of collaborative growth being able to explain incentives to merge in a conglomerative manner. Any firm not benefitting from it being part of a conglomerate would either split off or not join in the first place. Given our proposed transaction cost function in Equation 5, a conglomerate of size 20 experiences a per period drain of roughly 2.06×10^{-9} on their combined size by pooling their resources.

To examine the impact of mergers on the competitive landscape and to deepen our understanding of the role that transaction cost play, we fit a polynomial of degree two to the relation between conglomerate size and average market share of firms within a given conglomerate $\bar{\Omega}_{ct} = \beta_{0t} + \beta_{1t}|O_{ct}| + \beta_{2t}|O_{ct}|^2$ for each step in the model and display the result in Figure 3. We expect the conglomerate size to have a positive impact on the average market

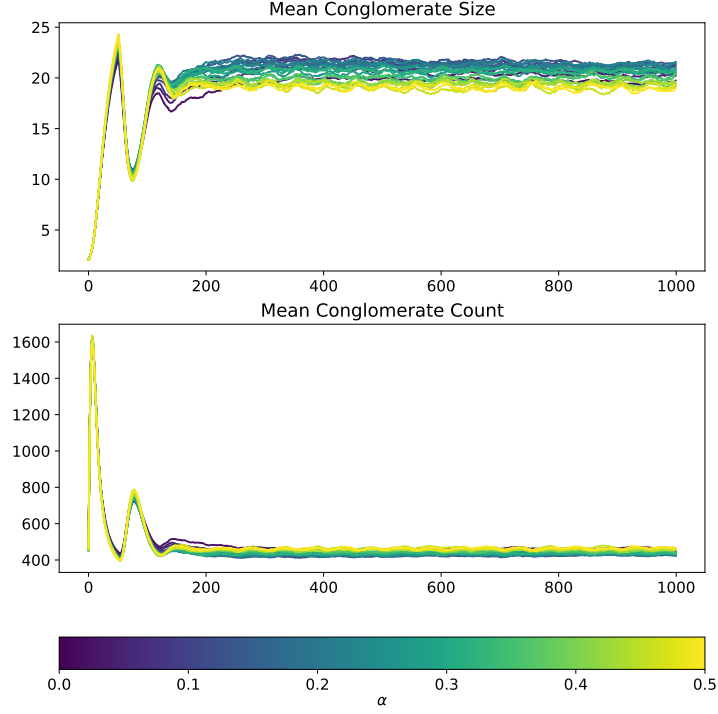


Figure 2: Conglomerate sizes and number of conglomerates.

share ($\beta_{1t} > 0$). However, we expect transaction costs to impose a drag high enough to negatively influence the average market share ($\beta_{2t} < 0$). For all levels of α these hypotheses hold once the initial fluctuations pass, that are due to the prevention of splitting during the initial look back period. Note that as expected β_{2t} is negative in the same region, meaning that growing too large will decrease competitiveness, thus highlighting the impact of transaction costs on the conglomerate size. Both of these variables initially increase in effect size before stabilizing asymptotically. Importantly, the scale of β_{1t} is many times larger than that of β_{2t} , meaning that even though the latter is multiplied by the square of conglomerate size, it would take conglomerates of sizes between 75 (for $\alpha = 0.02$) and 43 (for $\alpha = 0.5$) for conglomeration to reduce the average market share to 0. In our lowest sharing case these estimates predict a conglomerate of average size to have an average market share of 2.2 %, more than double the perfectly equal case. As we increase the sharing rate, these effects become ever more muted, pushing the estimates

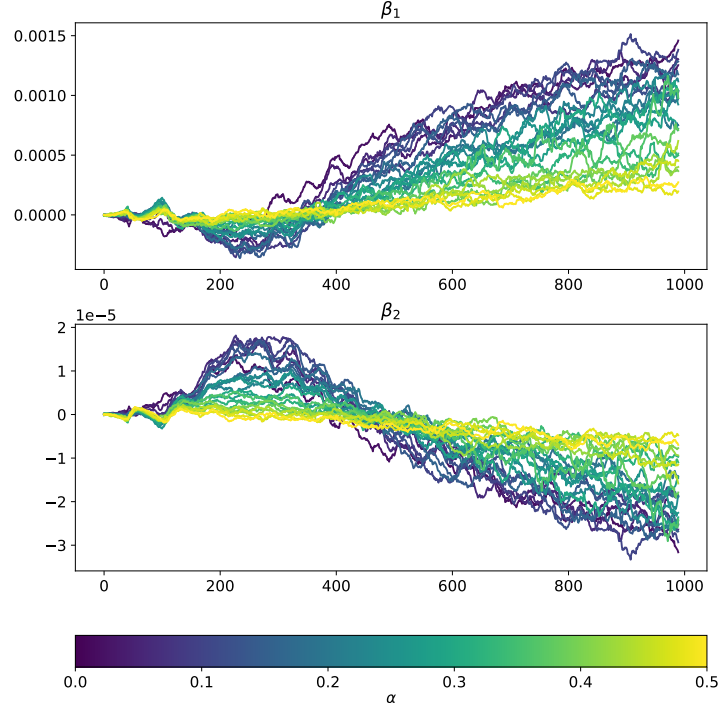


Figure 3: Average estimates for the impact of conglomerate size on market share.

for both β_{1t} and β_{2t} towards 0. This is a direct consequence of a decrease in the variability and scale of market share inequality.

Figure 4 provides an initial insight into this dynamic by illustrating the market share distribution across different quantiles. The graph displays the cumulative market share attained by each quantile of firms, ordered from largest to smallest. These are taken across every market and averaged across all quantiles for every point in time. In our setup with 100 firms per market a perfectly equal result would assign every firm a market share of 0.01. As expected, the case without pooling (the dashed line) returns the standard result of random multiplicative growth, exhibiting extreme inequality with the largest firm making up almost the entirety of the market and even the 90th percentile quickly dropping to 0% share. Increasing α just ever so slightly already changes this dynamic fundamentally for the largest firm, while allowing it to still capture market share. Furthermore, all quantiles become more equal in market share until they eventually remain close to

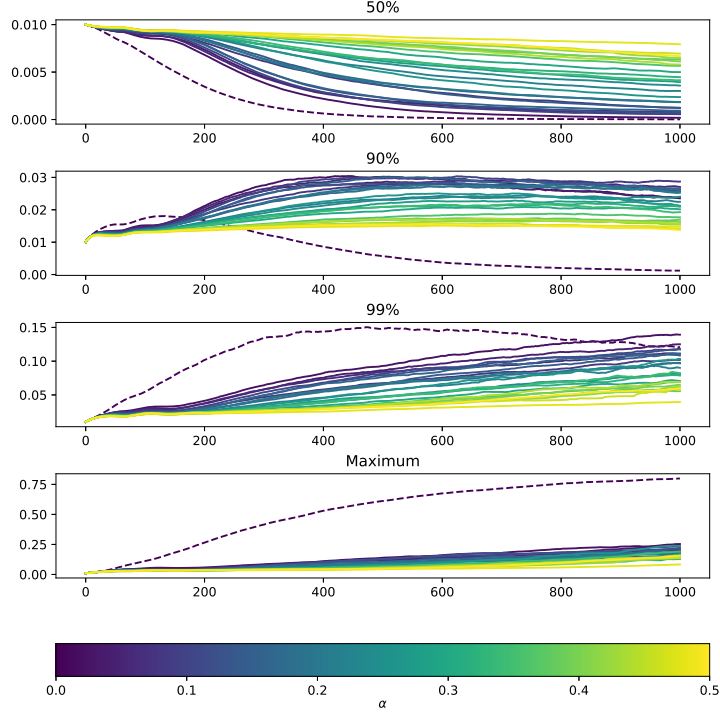


Figure 4: Market share of a given quantile.

equality for the entirety of the simulation. As already alluded to, this also explains why conglomerate size becomes ever less predictive with an increase in sharing as the variation within the market shares becomes ever smaller.

So far we have only looked at a static measure for a distribution, meaning that it can only provide information on the aggregate. To address this, we follow each firm throughout each simulation and tally both the difference between its maximum and minimum rank and the standard deviation of its ranks, and plot the resulting distributions over all firms and counterfactuals in Figure 5. Given that there are 100 firms per market, this sets an upper limit on mobility as measured by both metrics. The range shows that even in the no-pooling case the theoretical upper bounds of mobility are reached. However, we observe again that introducing just a small amount of sharing changes the dynamic drastically, as both measures shift completely in their pattern, raising mobility noticeably. In addition, increasing the sharing rate increases the variability of ranks, shifting the distribution to the right. In-

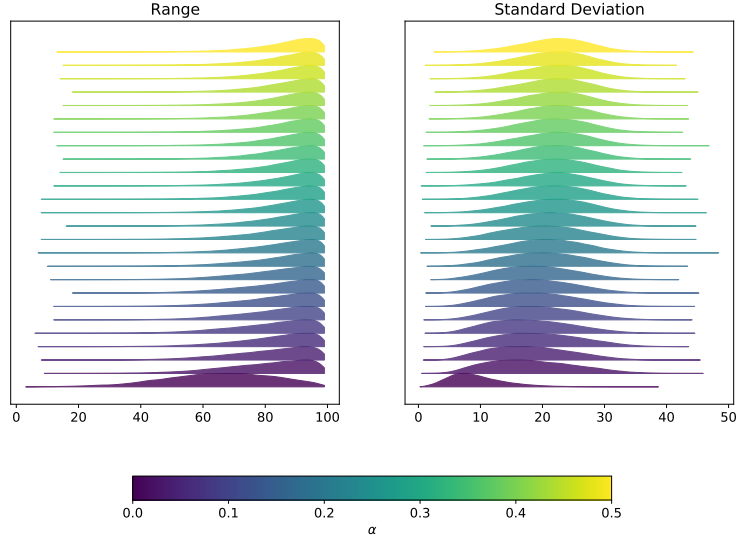


Figure 5: Distribution over ranks.

terestingly, given that the maximum range is already reached for any sharing level, we only see more mass of the distribution moving to the right. This means that with larger degrees of sharing more firms experience the chance to climb through the ranks. Similarly, while the no-sharing case displays only low levels of rank variation, increasing sharing seems to improve mobility greatly.

5 Discussion

A recurring theme in the search for determinants of conglomerate formation is diversification to mitigate risks. As specialization of the Smithian [27] and Ricardian [24] nature allows us to increase our returns to scale, and as we tend to increase in ever greater manner the degree of roundaboutness of production [31], the potential downsides from miscalculation or shocks looms ever larger. Mitigating these, however, is also becoming simpler [10], as both internal and external means of diversification [11] compete to be the most cost-effective insurance against adverse events. As demonstrated above, true diversification from conglomerate formation offers a reliable mitigation pathway and is applicable to many risk vectors.

Among these are macroeconomic or political shocks that are specific to certain geographic regions or sectors of the economy. The literature on international trade has picked up this theme arguing that cross-country foreign direct investment reduces country-specific risks [22] and even increases between less correlated markets [23]. Similarly, [2] argue that conglomeration can diversify across business cycles, while [3] proposes that it can even be used to match different product life cycles.

On the other hand, [14] offers a purely financial rational for conglomeration, where firms merge in order to mitigate financing frictions, and to provide internal capital markets [5]. In addition, pooling can be used to reduce the highly volatile returns on investments that are inherent to R&D efforts, or share the costs associated with them [10].

In fact, many of these effects are already being employed in the businesses seed financing of start-ups, where venture capital firms enforce a low pooling rate for their portfolio companies or those that are part of incubation batches in accelerator programs [6]. Interestingly, this concept is being extended to resources, other than financial, such as physical and human capital, and even reputation. It is thus evident, that pooling can be used to smooth any multiplicative processes, of which there are many that are of interest to economic inquiry [21, 26].

6 Conclusion

Pooling is not only beneficial to the conglomerate, as well as the constituent companies, but also to the competitive landscape as a whole, decreasing both static and dynamic inequality. We arrived at this result by showing that the standard perspective of the zero-sum nature of resource sharing misses critical details in a world of multiplicative growth. Secondly, we emphasize the difference between portfolio diversification and collaborative resource sharing, allowing us to go further than standard arguments of diversification. We find strong incentives to use conglomerate formation as an insurance mechanism, that if made widely available, would promote competition, from which especially new types of business models could benefit.

We see further possible refinements to our approach. Firstly, one could include a pricing mechanism that would allow for more far-reaching statements on the impact on consumers. This would come with a great deal of assumptions such as the nature and interaction of the prices, profits and the underlying growth dynamics. Beyond this, we are aware that the premise of normally distributed growth increments mischaracterizes the process of business growth [13]. In one respect, negative impacts on firm development tend to be heavy-tailed, meaning that impactful events occur much more frequently than normality would predict them [9, 30]. This also holds, though arguably to a lesser degree, for positive impacts, such as the strong growth potential observed in start-ups [18]. Given that large negative developments have an outsized impact on the overall performance, this seems like a strong use-case for the framework developed herein.

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